

Problem Set 6: Due on Tuesday, March 31, @noon

(100 points total)

1. **Full insurance for high enough discount factors (15 points).** Consider the version of Thomas and Worrall (1988) with limited commitment that we studied in class. That is, there is a risk-averse agent and a risk-neutral lender. The agent has preferences

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where u is strictly increasing, strictly concave, C^2 , and $\beta \in (0, 1)$. He faces a stochastic endowment stream $\{y_t\}_{t=0}^{\infty}$, which is i.i.d. over time, with $\Pr\{y_t = \bar{y}_s\} = \pi_s$ for $s = 1, \dots, S$. The agent can only borrow from the lender, while the lender can borrow or lend at the risk-free gross interest rate $R = \beta^{-1}$.

Show that there exists a threshold value of the discount factor, $\beta^* \in (0, 1)$, such that a full-insurance contract is self-enforcing for *both* parties after any possible history if and only if $\beta \geq \beta^*$.

2. **One-sided limited commitment with different discount factors (15 points).**

Consider the version of Thomas and Worrall (1988) with one-sided limited commitment that we studied in class, with the following modification. Suppose that the gross interest rate R at which the lender can save and borrow is strictly lower than β^{-1} .

- (a) (7 points) Suppose first that both parties can commit to long-term contracts. Characterize properties of the first-best allocations and the corresponding dynamics. Provide intuition for your results.
- (b) (8 points) Now suppose that the household can choose to walk away with its current income and live in autarky forever. How do the optimal contract and the corresponding dynamics look like in this case? Provide intuition for your results.

3. **Insurance with two-sided lack of commitment (25 points).** Consider an environment with two households, each with preferences given by

$$E \sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0, 1)$, $c_t \geq 0$ is consumption at time t , and the utility function $u(c)$ is strictly increasing, strictly concave, and C^2 . Assume that the endowments of the

households are perfectly negatively correlated. Whenever the first household receives $\bar{y}_s$, the second household receives $1 - \bar{y}_s$, that is, the total endowment always equals 1. Assume that y_t is i.i.d. over time with $\Pr\{y_t = \bar{y}_s\} = \pi_s$, $s = 1, \dots, S$, $\bar{y}_s < \bar{y}_{s+1}$, and $\bar{y}_s \in [0, 1]$. Assume also that π_s 's are such that the distribution of y_t is identical to that of $1 - y_t$.

The social planner has no access to neither borrowing nor lending opportunities, and is confined to reallocating consumption goods between the two households in each period. The social planner designs an optimal self-enforcing contract, which is subject to constraints that in each period after receiving its endowment, each household can choose to walk away and live in autarky from that period onward.

Denote the promised utility to household 1 by v , and the expected utility to household 2 from delivering (at least) v to household 1 by $P(v)$. Let $v_{\max}$ denote the highest possible value that can be promised to household 1. Denote consumption and the continuation utility to household 1 in state s by c_s and w_s , respectively. (In the optimal contract, they will depend on the promised utility v .)

- (a) (7 points) Set up a recursive problem corresponding to the optimal contract. Derive the first-order conditions and the envelope condition.
- (b) (8 points) Show that if the optimal allocation is different from autarky, then participation constraints of the two households cannot bind simultaneously.
- (c) (10 points) Show how the values to household 1 adjust (i.e., how w_s compares to v) depending on whether the participation constraints of household 1, household 2, or neither, bind. Provide intuition for your results.

4. **Computing Optimal Contracts with Limited Commitment (45 points).** Consider an economy with a risk-averse agent and a risk-neutral lender. The agent has preferences

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma},$$

where $\sigma = 2$ and $\beta \in (0, 1)$. He faces a stochastic endowment stream $\{y_t\}_{t=0}^{\infty}$, where y_t is i.i.d. over time, equal 1 or 2, both with probability 1/2. The agent can only borrow from the lender, while the lender can borrow or lend at the risk-free gross interest rate $R = \beta^{-1}$. In any period the agent can choose to walk away with his endowment and live in autarky forever. The lender can commit to long-term contracts.

- (a) (3 points) Write down the expression for the autarkic value to the household, v_{aut} , and compute it explicitly.

- (b) (7 points) Write down the problem of finding the optimal contract between the agent and the lender as a function of the initial promised value. Describe the dynamics implied by the optimal contract.
- (c) (14 points) Suppose that the agent's initial promised value is equal to his autarkic value, $v = v_{aut}$. Explicitly solve for the optimal contract as a function of the discount factor β . That is, you need to find $(\bar{c}_1, \bar{w}_1, \bar{c}_2, \bar{w}_2)$ as functions of β .
- (d) (5 points) For what value of β is $P(\bar{w}_2) = 0$ (where P denotes the lender's value function)?
- (e) (16 points) Write down an (implicit) expression for the threshold value β^* of the discount factor such that the optimal contract is also self-enforcing for the lender at any point in time if and only if $\beta \geq \beta^*$. Describe in words what this expression means. How does β^* compare to the value of β you found in part (d)? Explain the comparison intuitively.